ChainCred

Onchain Expertise Reputation Layer for Ethereum

github.com/chaincred
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GitHub contribution graph meets Ethereum — your wallet's expertise, provable onchain.

6	5	40+	8	7	20	103
CHAINS INDEXED	SCORE CATEGORIES	SCORING SIGNALS	SYBIL HEURISTICS	EXPERTISE BADGES	TRACKED PROTOCOLS	AUTOMATED TESTS

THE PROBLEM

Ethereum has no credible way to distinguish experienced builders from airdrop farmers, genuine DAO contributors from governance attackers, or real users from sybil accounts.

TOKEN GOVERNANCE
Wealth ≠ expertise

GITCOIN PASSPORT
Proves humanity, not skill

PORTFOLIO VALUE
Holding ≠ doing

POAPS / SOCIAL REP
Farmable, off-chain

CHAINCRED'S APPROACH

Index *what wallets do*, not what they hold. Score actions across 6 EVM chains, apply multiplicative sybil penalties, and publish results as verifiable EAS attestations.

1. **Expertise Score** (0–1000) — weighted composite of 5 onchain signal categories
2. **7 Badges** — specialist attestations (Builder, Governor, OG, Explorer, etc.)
3. **Sybil Resistance** — 8 heuristics that discount bots, not ban them
4. **Onchain Proof** — weekly Merkle root + EAS attestations for trustless verification

SCORING ENGINE — 5 CATEGORIES, 40+ SIGNALS

Builder	30%	deploys, chains, CREATE2, ERC-4337, verified source, longevity
Governance	25%	votes, DAOs, proposals, delegation, Safe multi-sig, reasoned votes
Temporal	20%	wallet age, bear market activity, consistency, entropy
Protocol Diversity	15%	unique protocols, chains, domain coverage, early adoption
Complexity	10%	calldata, failures, flashloans, permits, smart wallets

```
Final Score = (Builder × 0.30 + Governance × 0.25 + Temporal × 0.20 + Diversity × 0.15 + Complexity × 0.10) × Sybil Confidence
```

SYBIL RESISTANCE — DISCOUNT, NEVER BAN

8 multiplicative heuristics produce a confidence multiplier (0.0–1.0). Transparent and appealable.

-60% Cross-chain mirroring	-50% Funding clusters	-40% Temporal clustering	-30% Action repetition
-30% CEX freshness	-20% Zero failures	-15% Gas patterns	-15% MEV bot activity

7 EXPERTISE BADGES

Each badge maps to concrete onchain achievement — evaluated dynamically from indexed activity.

- **Builder** — 3+ deploys
- **Governor** — 5+ DAOs
- **OG** — pre-2020
- **Explorer** — 20+ protocols
- **Multichain** — 4+ chains
- **Trusted** — Safe signer
- **Power User** — depth + breadth

WHY IT'S BUILT THIS WAY

Actions Over Holdings
Token balances reward wealth. ChainCred only indexes *transactions* — deployments, votes, protocol interactions. You can't buy a score.

Multi-Signal Categories
Each category uses 4–9 capped signals. Over-capping prevents any single signal from dominating — no wallet can max all signals.

Multiplicative Sybil Penalties
Penalties compound: 3 flags = 0.6 × 0.7 × 0.4 = 0.168 confidence. Each extra flag bites exponentially.

Bear Market = High Signal
Activity during downturns is the hardest signal to fake retroactively. Cross-cycle persistence is near-unforgeable proof of genuine participation.

Trustless Verification
Weekly Merkle roots onchain let anyone verify a score without trusting ChainCred. Combined with EAS attestations — composable by any protocol.

Fail-Open Everything
If Etherscan, RPC, or Redis is down, scoring still works. Enrichments degrade gracefully — no external dependency blocks the core pipeline.

KEY FEATURES

REST + WebSocket API
21 endpoints: scores, badges, sybil, leaderboard, timeline, proofs, webhooks, appeals

SVG Score Cards
Shareable cards with category bars, badges, ENS names. OG + Farcaster meta tags.

Verification Widget
Web Component (Shadow DOM). Gate anything by score + badges.

Appeals + Webhooks
Sybil appeals with score floor. HMAC-signed webhook delivery.

ARCHITECTURE

HyperSync (6-chain indexer) → **PostgreSQL** (wallet activity) → **Scoring Engine** (5 categories + sybil) → **Hono API** (REST + WS + Redis)

Merkle Tree (weekly generation) → **Solidity Contracts** (Registry + Merkle + Resolver) → **EAS Attestations** (composable onchain)

Bun · TypeScript · Hono · PostgreSQL · Redis · Solidity 0.8.24 · Foundry · SvelteKit · HyperSync · EAS · viem · Docker

Bun monorepo • 6 packages • ~3,600 LOC • 103 tests • MIT License